NETGRAPH — Mining Hidden Relationships Between Skills, Technologies & Career Opportunities

Abstract: NETGRAPH — Mining Hidden Relationships Between Skills, Technologies & Career Opportunities is a data mining and data wrangling system designed to analyze complex job-market data and uncover meaningful relationships between skills, technologies, and job roles. The system begins by collecting and preprocessing raw data through data wrangling techniques such as data cleaning, duplicate removal, missing-value handling, skill normalization, and transformation into a structured format. The processed data is then analyzed using data mining techniques, including clustering, classification, anomaly detection, relationship analysis, and graph-based analysis, to identify frequently associated skills, important technologies, similar job profiles, communities, and unusual patterns. These relationships are represented as nodes and connections within an interactive network, enabling users to search for a specific skill or technology and explore its related skills, technologies, and career opportunities. The system primarily uses **Python, Pandas, NumPy, and Scikit-learn**, with visualization techniques to present network structures, patterns, rankings, and analytical results in an understandable manner. By combining data wrangling, data mining, network analysis, and interactive visualization, NETGRAPH transforms large and complex career datasets into meaningful insights, helping users understand how different skills and technologies are connected and how these relationships are distributed across various job roles.